Data Sheet

EXPERIMENT NO. 10

Title: - Simulation of Phase Shift Oscillator using Op-Amp — Study of Frequency of Oscillation by Varying Component Values (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1	Op-Amp	IC 741	1
2	Resistor	10 kΩ	4
3	Feedback Resistor	330 kΩ	1
4	Capacitors	0.1 μf/0.01 μf/0.001 μf	3
5	DC Power Supply	15 V	1
6	Oscilloscope.	—	1

Important Formulae: -

Barkhausen Criterion for Sustained Oscillation: -

Loop Gain, $|Av \cdot \beta| = 1$ and total phase shift around the loop = 360° (or 0°)

- The op-amp (inverting amplifier) contributes 180° phase shift
- The three-section RC feedback network contributes the remaining 180° (60° per section) at the frequency of oscillation

Frequency of Oscillation (three equal RC sections): -

$f = 1 / (2\pi RC \sqrt{6})$

- R = resistance and C = capacitance of each RC section (all equal)

Minimum Gain Condition for Oscillation: -

Feedback factor of a 3-section RC network, $\beta = 1/29$

Hence the amplifier must provide a gain, $|Av| = Rf / R1 \geq 29$

Observation Table: -

Sl. No.	R (kΩ)	C (nF)	Rf (kΩ)	R1 (kΩ)	f (Theoretical) Hz	f (Practical / Measured) Hz

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Title :- Simulation of phase Shift Oscillator using OP-Amp - Study of frequency of oscillation by varying Component values (using Tinkercad / proteus).

Theory :-

⊙ Barkhausen Criterion for Sustained Oscillation: Loop Gain $|Av.\beta| = 1$ and total Phase Shift around the loop = 360°.

⊙ The OP-Amp (inverting amplifier) contributes a 180° phase shift.

⊙ The three section RC feedback network contributes the remaining 180° at the frequency of Oscillation.

⊙ Frequency of Oscillation:

$f = 1/(2\pi RC\sqrt{6})$. R = Resistance, C = Capacitance.

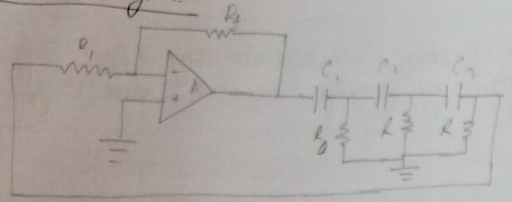
⊙ Minimum Gain Condition for Oscillation: Feedback factor of a 3-section RC network, $\beta = 1/29$. Hence the amplifier must provide a gain $|Av| = Rf/R_1 \geq 29$.

Component Required :-

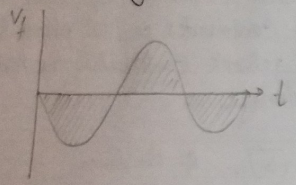
	Quantity
1. OP-Amp (IC 741)	1
2. Resistors (10 kΩ)	4
3. Feedback Resistor (330 kΩ)	1
4. Capacitor (0.1 μf/ 0.01 μf/0.001 μf)	3
5. DC power Supply (15 V)	1
6. Oscillo scope (Virtual)	1

Teacher's Signature

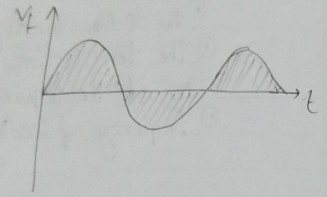
Circuit Diagram:-

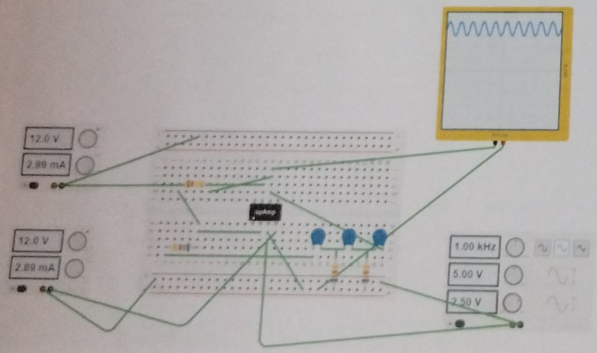


Input Signal



Output Signal





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Observation Table:-

SN	R (KΩ)	C (μf)	Rf (KΩ)	R1 (KΩ)	f (Theoretical) Hz	f (Practical) Hz
1.	10	0.1	330	10	65 Hz	64.2
2.	10	0.01	330	10	650 Hz	645.2
3.	10	0.01	330	10	1.38KHz	1.37
4.	10	0.001	330	10	6.50 KHz	6.42

Calculation:-

① $R = 10K\Omega = 10000\ \Omega$

① $C = 0.1\mu f = 0.1 \times 10^{-6} F$

① Practical Frequency = 64.2 Hz

$f(\text{Theoritical}) = 1/(2\pi RC\sqrt{6})$

$= 1/(2 \times 3.1416 \times 10000 \times 0.1 \times 10^{-6} \times 2.449)$ Hz

$= 64.97$ Hz

$\%\ \text{Error} = \left|\frac{f(\text{the}) - f(\text{Prac})}{f(\text{the})}\right| \times 100 = 1.23\ \%$

Conclusion:-

The phase shift oscillator simulation successfully generated a continuous sine wave, confirming that the circuit meets the Barkhausen criterion by providing sufficient gain ($R_f/R_1 \geq 29$) and a total phase shift of 360°. By varying the capacitor values, it was observed that the frequency of oscillation is inversely proportional to the capacitance and the practically measured frequencies closely matched the theoritical formula ($f = 1/2\pi RC\sqrt{6}$) with minimal error.

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